NATIONAL OPEN UNIVERSITY OF NIGERIA
FACULTY OF SCIENCE
DEPARTMENT OF COMPUTER SCIENCE
2023_2 EXAMINATIONS

COURSE CODE: CIT 383
COURSE TITLE: INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING
CREDIT: 2 UNITS
TIME ALLOWED: 2 HOURS
INSTRUCTION: ANSWER QUESTION 1 AND ANY TW0 (2) OTHERS

QUESTION ONE. (30 Marks)

a. Create a non-executable class that has a private instance variable that can hold the first name of a student. Your class should contain methods to assign values to and retrieve values from the instance variable. ***(8 Marks)***
b. Differentiate between a default constructor and a no argument constructor? ***(7 Marks)***
c. Explain the concept behind garbage collection. ***(8 Marks)***
d. Using a schematic diagram differentiate between overridden and inheritance? ***(7 Marks)***

QUESTION TWO. (20 Marks)

a. Differentiate between an abstract class and a concrete class. ***(6 Marks)***
b. Describe what is meant by conversion operator overloading ***(6 Marks)***
c. Write a program that will overload a conversion operator. ***(8 Marks)***

QUESTION THREE. (20 Marks)

a. Give four properties of abstract data type (ADT) ***(6 Marks)***
b. Discuss concisely on the types of polymorphism? ***(8 Marks)***
c. List six (6) examples of passing message style. ***(6 Marks)***

QUESTION FOUR. (20 Marks)

a. What is the difference between a Class and an Object? ***(8 Marks)***
b. Write a program that computes the Area of a sphere. ***(12 Marks)***

QUESTION FIVE. (20 Marks)

a. Discuss four (4) benefits of bundling code into individual software objects? ***(12 Marks)***
b. Discuss concisely encapsulation and a program to buttress your explanation. ***(8 Marks)***